

Olympiad Test Paper — Mathematics (Class 7) — Paper 5

Chapter 13: Connecting the Dots

Paper 5 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	13 — Connecting the Dots
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. These problems stack constraints: solve for a missing value only after fixing the total, resort before reading any median, weight every combined mean by its group size, never average two means naively, treat the change in a mean as distinct from the new data point, read a bar's true value from its scale and axis-start (not its apparent height), and for probability count favourable outcomes over equally likely total outcomes. Do not write on this paper — mark answers on your answer sheet.

1. The mean of seven numbers is 30. The mean of the smallest four of them is 24 and the mean of the largest four is 39. What is the value of the fourth number (which belongs to BOTH groups, since the two groups of four overlap in exactly one number)?

- (A) 30 (B) 63
(C) 15 (D) 42

2. The mean of 9 numbers is 14. A 10th number is added and the mean rises to 16; then an 11th number is added and the mean drops back to 15. What is the 11th number?

- (A) 15 (B) 36
(C) 5 (D) 9

3. Every value in a data set of 6 numbers is first multiplied by 4 and then 5 is added to each result. If the original mean was 7, the new mean is:

- (A) 33 (B) 28
(C) 48 (D) 13

4. Class A (mean 48) and Class B (mean 72) are merged and the combined mean is exactly 54. The ratio of the number of students in Class A to Class B is:
- (A) 1 : 3 (B) 3 : 1
(C) 1 : 1 (D) 2 : 1
5. Seven numbers in increasing order are 5, 8, 12, x , 26, 30, 33. The median is required to equal half the range. The value of x is:
- (A) 14 (B) 28
(C) 7 (D) no valid x exists
6. Eight numbers in increasing order are 4, 9, 13, p , q , 24, 28, 31 with $p < q$ and $p + q = 36$. The data also has a unique mode of 24 (so $q \neq 24$ and exactly one extra copy idea is not needed here — just $q < 24$). Which pair (p , q) fits the order and the sum?
- (A) $p = 19$, $q = 17$ (B) $p = 16$, $q = 18$
(C) $p = 17$, $q = 19$ (D) $p = 15$, $q = 22$
7. A data set is 2, 2, 5, 5, 5, 8, 8, 9. The fewest single-value edits (each edit changes one entry to any value) needed to make the data have EXACTLY two modes, namely 2 and 8, is:
- (A) 2 (B) 1
(C) 3
(D) 0 — it is already bimodal with modes 2 and 8
8. Two players each batted 6 innings with the same mean of 35. Pari's scores have range 18; Ria's range 54. A commentator says 'equal means show equal ability.' The mathematically sound comparison is:
- (A) Ria is better because a larger range means she can score very high
(B) Their totals are equal (both 210), but Pari is the more consistent batter because her smaller range (18) shows her scores cluster more tightly around the mean than Ria's (range 54)
(C) They are identical players since their means match
(D) Pari is weaker because a small range means all her scores are low
9. Five positive whole numbers have mean 9, median 8 and a unique mode of 6 (appearing more often than any other value). Which set meets ALL three conditions?
- (A) 6, 6, 8, 10, 15 (B) 6, 6, 8, 8, 17
(C) 6, 7, 8, 9, 15 (D) 6, 6, 7, 10, 16
10. A data set of 9 values is 3, 5, 7, 9, 11, 13, 15, 17, 19 (mean 11, median 11). The largest value 19 is replaced by 119. Which statement is EXACTLY correct?
- (A) Both the mean and the median rise to about 22.1
(B) The mean stays 11 while the median rises

- (C) Neither changes, because only one value moved
(D) The mean rises to about 22.1 while the median stays 11

11. The mean of 12 numbers is 20. Three numbers — 8, 12 and 16 — are removed and replaced by two numbers whose sum is 50. The new mean (of the now 11 numbers) is:

- (A) 20 (B) About 23.1
(C) About 21.2 (D) 50

12. The mean of 15 observations was computed as 24, but later TWO entries were found wrong: 30 had been recorded as 13, and 9 had been recorded as 41. The corrected mean is:

- (A) 24 (B) 25
(C) 23 (D) 22

13. A pie chart of 360° shows how a family's monthly budget of ₹24000 is split: Food 120° , Rent 150° , Savings 60° , Other 30° . How much MORE is spent on Rent than on Savings?

- (A) ₹4000 (B) ₹10000
(C) ₹90 (D) ₹6000

14. Adding 9 to each number of a data set produces a new mean of 31, and the original total of the data was 110. The number of values, n , is:

- (A) 5 (B) 10
(C) 4 (D) 22

15. Two pictographs use DIFFERENT scales. In graph 1 one symbol = 18 marbles; in graph 2 one symbol = 12 marbles. Jar A shows $3\frac{1}{2}$ symbols on graph 1 and Jar B shows 5 symbols on graph 2. Which jar has more marbles and by how many?

- (A) Jar B, by $1\frac{1}{2}$ marbles (B) they are equal
(C) Jar A, by 3 marbles (D) Jar B, by 18 marbles

16. Three groups have means 8, 15 and 24. The first group has 3 numbers and the third has 4 numbers. If the overall mean of all the numbers is 16, how many numbers are in the second group?

- (A) 3 (B) 8
(C) 5 (D) 2

17. Class P has 40 students with mean score 65. After merging with Class Q (mean 85) the combined mean is 77. How many students are in Class Q?

- (A) 20 (B) 40
(C) 60 (D) 30

18. Eight numbers in increasing order are 5, 8, 12, a , b , 21, 25, 30 with $a < b$. Their median is 17. Which pair (a , b) is IMPOSSIBLE?

- (A) $a = 15$, $b = 19$ (B) $a = 16$, $b = 18$
(C) $a = 14$, $b = 20$ (D) $a = 18$, $b = 16$

19. Seven numbers written in increasing order are 5, 8, x , 14, 18, 18, 24. The data has mean 14 and a unique mode of 18. The value of x is:

- (A) 14
- (B) 11
- (C) 12
- (D) 9

20. In a data set every value appears exactly once except one value v that appears FOUR times. Three copies of a new value w ($w \neq v$, previously absent) are added. After this, the mode is:

- (A) w , the new unique mode
- (B) both v and w , a tie at four each
- (C) still v alone, because v appears four times while w reaches only three — fewer than v 's four, so the set stays unimodal at v
- (D) no mode at all

21. A double bar graph shows a clinic's patients across two years. Children: 2023 $\rightarrow$ 90, 2024 $\rightarrow$ 126. Adults: 2023 $\rightarrow$ 150, 2024 $\rightarrow$ 132. Which single statement is fully correct?

- (A) Both groups grew because both 2024 bars are tall
- (B) Children overtook Adults in 2024
- (C) Adults grew faster than Children because their bars are taller overall
- (D) Children rose by 36 while Adults fell by 18, yet Adults' 2024 count (132) still exceeds Children's (126)

22. A bag holds 5 red, 3 green and 2 blue balls, identical except for colour. One ball is drawn at random. The probability that the ball is NOT green is:

- (A) $3/10$
- (B) $7/10$
- (C) $1/2$
- (D) $2/10$

23. A list of five distinct whole numbers has median 15. The two largest are each increased by 8 and the smallest is decreased by 5, with the order preserved. The new median is:

- (A) 23
- (B) 10
- (C) 15
- (D) 20

24. Group P (mean 24) has three times as many numbers as Group Q (mean 60). The mean of all the numbers combined is:

- (A) 42
- (B) 33
- (C) 36
- (D) 28

25. The mean of six numbers is 25. Replacing the smallest number, which is 12, by an unknown value w makes the new mean 29. The value of w is:

- (A) 36
- (B) 29
- (C) 24
- (D) 41

26. The mean of 10 numbers was computed as 60. Later it was found that a value of 95 had been entered as 59, and a value of 28 had been entered as 82. The corrected mean is:

- (A) 60 (B) 61.8
(C) 57 (D) 58.2

27. A list of seven distinct numbers has median 30. The median value 30 is itself removed, leaving six numbers. The new median is:

- (A) the average of the two numbers that had been immediately on either side of 30 (the original 3rd and 5th values)
(B) still 30 (C) the original 3rd value alone
(D) the original 5th value alone

28. A set of 12 numbers has mode 6 appearing exactly 5 times and value 9 appearing twice. Three copies of 9 and one extra copy of 6 are added. After the additions, the mode is:

- (A) 9, appearing 5 times (B) both 6 and 9, a tie
(C) 6, appearing 6 times (more than 9's total of 5)
(D) there is no mode

29. The mean of 8 numbers is 30. A 9th value equal to 30, a 10th value equal to 10 and an 11th value equal to 50 are added. The final mean of all 11 numbers is:

- (A) 28 (B) About 31.8
(C) 30 (D) 33.3

30. A magazine's bar graph has a vertical axis starting at 500. Region X's bar (value 530) appears about three times as tall as Region Y's bar (value 510). The honest conclusion about the ratio of the actual values is:

- (A) X is about three times Y, as the bars show (B) Y is essentially zero
(C) X is only about 1.04 times Y; the truncated axis turned a 20-unit gap into a threefold visual illusion
(D) the graph cannot be read at all

31. The mean of a data set is 14. Each value is replaced by $(5 \times \text{the value} - 12)$. The mean of the new set is:

- (A) 70 (B) 58
(C) 14 (D) 22

32. A double bar graph compares two libraries' membership in two languages. Library X: Hindi 45, English 30. Library Y: Hindi 18, English 52. Which single statement is fully correct?

- (A) Library Y is larger overall and has the smaller gap
(B) Library X is larger overall (total 75 vs Y's 70), yet Y has the larger language gap ($|18 - 52| = 34$ against X's $|45 - 30| = 15$)
(C) Both libraries have equal totals (D) Library X has the larger language gap

33. A spinner is divided into 8 equal sectors numbered 1 to 8. It is spun once. The probability that the number landed on is a factor of 6 is:

- (A) $\frac{3}{8}$
- (B) $\frac{1}{4}$
- (C) $\frac{5}{8}$
- (D) $\frac{1}{2}$

34. The mean of three numbers is 16 and the mean of five other numbers is 24. A single extra number x is included with all eight; the grand mean of all nine numbers becomes 22. The value of x is:

- (A) 22
- (B) 168
- (C) 18
- (D) 30

35. A data set of 6 numbers has mean 14 and median 12. The largest value, originally 22, is increased to 40. After this change, which is GUARANTEED?

- (A) The mean increases by 3 (to 17) while the median stays 12
- (B) The mean and median both increase
- (C) The mean stays 14 while the median increases
- (D) Neither the mean nor the median changes

36. A double bar graph compares boys and girls in two clubs. Drama: boys 21, girls 13. Robotics: boys 8, girls 30. Which single statement is fully correct?

- (A) Drama is more popular overall and has the bigger gap
- (B) Robotics is more popular but Drama has the bigger gap
- (C) Both clubs have equal totals so neither is more popular
- (D) Robotics' total (38) exceeds Drama's (34), so Robotics is more popular; and Robotics' gender gap $|8 - 30| = 22$ exceeds Drama's gap $|21 - 13| = 8$

37. A data set of 6 numbers has mean 18 and median 16. The two smallest values, originally 7 and 9, are both replaced by 3. After the change, which is EXACTLY true?

- (A) The mean stays 18 and the median stays 16
- (B) The mean falls to about 16.3 while the median stays 16
- (C) Both the mean and the median fall by 2
- (D) The mean falls and the median also falls to 12

38. A data set of 5 numbers has mean 22 and median 22. The two largest values are each increased by 9 and the two smallest are each decreased by 9, the order preserved. Which is EXACTLY true after the change?

- (A) The mean stays 22 and the median stays 22
- (B) The mean rises by 9 and the median stays 22
- (C) The mean stays 22 but the median rises
- (D) Both fall by 9

39. The mean of 5 numbers is 20. Each number is increased by twice its own position number (1st by 2, 2nd by 4, 3rd by 6, 4th by 8, 5th by 10). The new mean is:

- (A) 20 (B) 30
(C) 22 (D) 26

40. A list of 9 numbers has range 60. The largest value is decreased by 15 and the smallest is increased by 9, the order otherwise unchanged. The new range is:

- (A) 60 (B) 84
(C) 45 (D) 36

41. The mean of 16 numbers was found to be 35, but a single value was entered as 23 when it should have been 87. The corrected mean is:

- (A) 35 (B) 39
(C) 37 (D) 43

42. A list of six numbers in increasing order is 10, 14, 18, 22, 26, 30. If the value 10 is removed, what happens to the median?

- (A) It rises from 20 to 22 (B) It stays at 20
(C) It falls to 18 (D) It rises to 24

43. A bar graph (scale 1 cm = 6 marks) shows five students' bars of 4 cm, 5 cm, 6 cm, 5 cm and 5 cm. A sixth student scored exactly the mean of those five. The sixth student's bar height would be:

- (A) 5 cm (B) 30 cm
(C) 25 cm (D) 1 cm

44. Group A of 5 numbers has mean 12; Group B of 3 numbers has mean 20; Group C of 2 numbers has mean 30. The mean of all 10 numbers is:

- (A) 18 (B) About 20.7
(C) 62 (D) 15

45. The marks of nine students are 7, 11, 11, 11, 14, 18, 20, 21, 22. Compute (mean + mode – 2 × median).

- (A) 0 (B) –2
(C) 2 (D) 15

46. The marks of seven students are 14, 14, 14, 18, 22, 22, 26. The correct increasing order of mean, median and mode is:

- (A) mean < median < mode (B) median < mean < mode
(C) mean = median = mode (D) mode < median < mean

47. The mean of a data set is 40. Each value is first halved, then 12 is added to each result. The mean of the new set is:

- (A) 20 (B) 32
(C) 52 (D) 26

48. Two pictographs use DIFFERENT scales: in graph 1 one symbol = 25 trees; in graph 2 one symbol = 20 trees. Forest M shows 3 symbols on graph 1; Forest N shows 4 symbols on graph 2. Which forest has more trees, and by how many?

- (A) Forest N, by 5 trees (B) Forest M, by 1 tree
(C) they are equal (D) Forest N, by 1 tree

49. The mean of 14 numbers was found to be 50, but a single value was entered as 96 when it should have been 18. The corrected mean is:

- (A) 50 (B) About 44.4
(C) About 55.6 (D) 47

50. Nine numbers in increasing order have median 50 (the 5th value). The four LARGEST values (positions 6 to 9) are each increased by 30, and the four SMALLEST (positions 1 to 4) are each decreased by 30. The new median is:

- (A) 80 (B) 20
(C) 65 (D) 50 (unchanged)

51. A cricketer's mean over 12 innings is 38. He wants to raise his mean to 42 over his next two innings combined (so 14 innings in all). How many runs must he score across those two innings together?

- (A) 84 (B) 120
(C) 132 (D) 66

52. A data set of 8 numbers has mean 25. Four of the numbers are each equal to the mean. The other four have a sum S. Then S equals:

- (A) 200 (B) 25
(C) 50 (D) 100

53. The mean of five numbers is 40. The smallest is 12 and the largest is 84. The middle three numbers are all equal to one another. Each of those middle numbers equals:

- (A) 40 (B) About 34.7
(C) 104 (D) 52

54. A pie chart of a school's 720 students by favourite sport shows Cricket 140° , Football 100° , Hockey 80° , Other 40° . How many students chose Football?

- (A) 100 (B) 280
(C) 200 (D) 160

55. A double bar graph shows two shops' sales (in ₹ thousand) over three quarters. Shop P: 40, 70, 10 (total 120). Shop Q: 30, 30, 60 (total 120). A reporter writes 'identical sales.' The most accurate correction is:

- (A) Their totals match (120 each), but the quarter-by-quarter pattern differs — P peaks in quarter 2 while Q peaks in quarter 3 — so the sales are not identical period by period
(B) The totals differ, so the reporter is right to call them different

(C) They are identical because the totals are equal

(D) Shop Q sold more because its highest single quarter (60) is the largest value

56. The mean of 9 numbers is 22. If the largest number, 46, is replaced by a value that makes the mean a whole number 20, the replacement value is:

(A) 18

(B) 46

(C) 28

(D) 20

57. A set of data has mean 50, median 50 and mode 50. A single new value of 130 is added. Which statement is GUARANTEED correct?

(A) All three rise to about the same new value

(B) The mode jumps to 130 because 130 is the new largest value

(C) The mean rises above 50, while the mode stays 50 and the median rises by at most a little (it does not jump to 130)

(D) Nothing changes because the data was already balanced

58. A list of six numbers 5, 9, 13, 13, 17, y has mode 13 and median 13. Which value of y keeps BOTH true?

(A) $y = 11$, which would make the median 13 but the mode would fail

(B) $y = 5$, giving median 11 (the mode 13 survives but the median fails)

(C) y must be exactly 18 for both to hold

(D) $y = 13$ — then 13 occurs three times (so the mode is 13) and the 3rd and 4th sorted values are both 13 (so the median is 13)

59. A magazine graph with a vertical axis starting at 1000 shows City X at 1040 looking far taller than City Y at 1010. A reporter writes 'City X is much bigger than City Y.' The mathematically honest statement is:

(A) The two cities differ by only 30 units (1040 vs 1010, about 3%); the truncated axis starting at 1000 makes a tiny gap look huge

(B) City X is about four times City Y (C) City Y is essentially zero

(D) the graph proves City X is much bigger

60. A standard die has faces 1 to 6. It is rolled once. Let A be the event 'an even number' and B be the event 'a number greater than 4'. The probability of getting a number that is in A OR in B (an even number or a number greater than 4) is:

(A) $3/6$

(B) $5/6$

(C) $4/6$

(D) $2/6$

Solutions — Mathematics (Class 7), Chapter 13: Connecting the Dots — Paper 5

Paper 5 — (progressively harder)

Marking: +4 correct, −1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	11	B	21	D	31	B	41	B	51	C
2	C	12	C	22	B	32	B	42	A	52	D
3	A	13	D	23	C	33	D	43	A	53	B
4	B	14	A	24	B	34	D	44	A	54	C
5	A	15	C	25	A	35	A	45	B	55	A
6	C	16	B	26	D	36	D	46	D	56	C
7	A	17	C	27	A	37	B	47	B	57	C
8	B	18	D	28	C	38	A	48	A	58	D
9	A	19	B	29	C	39	D	49	B	59	A
10	D	20	C	30	C	40	D	50	D	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (D) All seven total $7 \times 30 = 210$. The four smallest total $4 \times 24 = 96$; the four largest total $4 \times 39 = 156$. Because both groups of four are taken from only seven numbers, together they list $4 + 4 = 8$ positions but there are only 7 numbers, so exactly one number is counted twice — the overlapping fourth value. Hence $96 + 156 = 252$ counts every number once plus that shared value an extra time: $252 = 210 + (\text{shared value})$, giving the shared value $= 252 - 210 = 42$. Copying the overall mean gives 30; adding the two group totals' difference gives 63; halving the overlap idea gives 15. The overlap value is 42.

2. (C) Nine numbers total $9 \times 14 = 126$. After the 10th, total $= 10 \times 16 = 160$, so the 10th $= 160 - 126 = 34$. After the 11th, the total must be $11 \times 15 = 165$, so the 11th $= 165 - 160 = 5$. The value 34 is the 10th number, not the 11th; 15 merely copies the final mean; 9 is the original count. Only 5 pulls the average back to 15.

3. (A) A transform of the form new $= 4 \times \text{old} + 5$ carries straight to the mean: new mean $= 4 \times 7 + 5 = 28 + 5 = 33$. Stopping after multiplying gives 28; multiplying the already-

shifted value (e.g. $4 \times (7 + 5)$) gives 48; treating the transform as $\text{old} + 4 + 5$ gives 16-ish, and $13 = 7 + 5 + 1$ ignores the scaling. The new mean is 33.

4. (B) Let A have a students and B have b . Then $(48a + 72b) \div (a + b) = 54$, so $48a + 72b = 54a + 54b$, giving $18b = 6a$, i.e. $a : b = 18 : 6 = 3 : 1$. Because the combined mean 54 sits much closer to A's 48 than to B's 72, the lower-mean class A must be the larger group, matching 3 : 1. A 1 : 3 split would pull the mean toward 72; 1 : 1 gives mean 60; 2 : 1 gives mean 56.

5. (A) For seven sorted values the median is the 4th value, x , which must satisfy $12 \leq x \leq 26$. The range $= 33 - 5 = 28$, so half the range $= 14$. Setting the median equal to 14 forces $x = 14$, and since $12 \leq 14 \leq 26$, the value 14 legitimately occupies the 4th sorted slot ($8 < 12 < 14 < 26$ stays increasing). So $x = 14$ works. Taking 28 mistakes the range for the median; 7 halves 14 again; and a valid x certainly does exist since 14 lies in the allowed interval.

6. (C) For eight sorted values, p and q occupy positions 4 and 5, so they must satisfy $13 \leq p < q \leq 24$ and $p + q = 36$. Among the options, $p = 17, q = 19$ gives $17 + 19 = 36$, respects $p < q$, and fits $13 < 17 < 19 < 24$ ✓. The pair $(19, 17)$ violates $p < q$; $(16, 18)$ sums to 34, not 36; $(15, 22)$ sums to 37, not 36. Only $(17, 19)$ satisfies both the order and the sum.

7. (A) Currently 5 appears three times, 2 twice, 8 twice, 9 once — so 5 is the unique mode (not bimodal). To make 2 and 8 the only modes, each must be the most frequent and tie each other, while 5 must not exceed them. One clean route: change one 5 into a 2 (now 2 appears three times) and change the other-needed step — change another 5 into an 8 (now 8 appears three times), leaving 5 with only one copy. That is two edits, giving 2,2,2,5,8,8,8,9: modes 2 and 8 each three times. One edit alone cannot both knock 5 below and raise both 2 and 8 to a tie. So two edits are needed; it is not already bimodal (5 leads), and three edits are more than necessary.

8. (B) Mean 35 over 6 innings gives each a total of $6 \times 35 = 210$, so totals match — equal means cannot, by themselves, distinguish the players. Range measures spread: Pari's 18 keeps her innings in a narrow band while Ria's 54 swings widely. A smaller spread about the same mean signals greater consistency, so Pari is steadier. A large range is not proof of higher scoring, equal means do not make players identical, and a small range describes tight clustering, not uniformly low scores.

9. (A) Need total $5 \times 9 = 45$, the 3rd sorted value $= 8$, and 6 the unique most frequent value. Test 6, 6, 8, 10, 15: sum $= 45$ ✓, 3rd value $= 8$ ✓, 6 appears twice and every other value once, so 6 is the unique mode ✓. The set 6, 6, 8, 8, 17 has 6 and 8 each appearing twice (bimodal), so 6 is not unique. The set 6, 7, 8, 9, 15 has no repeated value, so it has no mode. The set 6, 6, 7, 10, 16 has its 3rd value $= 7$, so its median is 7, not 8. Only 6, 6, 8, 10, 15 satisfies every condition.

10. (D) New total = (old total) – 19 + 119 = $(9 \times 11) - 19 + 119 = 99 + 100 = 199$, so the new mean = $199 \div 9 \approx 22.1$ (up about 11). The sorted list 3, 5, 7, 9, 11, 13, 15, 17, 119 still has 11 as its 5th (middle) value, so the median is unchanged at 11. This is the outlier signature: the mean is dragged by an extreme value while the median resists. Claiming the median also jumps ignores its robustness; saying the mean stays 11 ignores the changed total; replacing a value certainly changes the mean.

11. (B) Original total = $12 \times 20 = 240$. Removing 8, 12, 16 (sum 36) and adding two numbers summing 50 gives a new total of $240 - 36 + 50 = 254$. The count changes from 12 to $12 - 3 + 2 = 11$, so the new mean = $254 \div 11 \approx 23.1$. Choosing 20 copies the old mean; dividing 254 by 12 (the wrong count) gives ≈ 21.2 ; 50 mistakes the inserted sum for the mean. The new mean is about 23.1.

12. (C) Recorded total = $15 \times 24 = 360$. First error: 13 should be 30, so add 17. Second error: 41 should be 9, so subtract 32. Net correction = $+17 - 32 = -15$. True total = $360 - 15 = 345$, mean = $345 \div 15 = 23$. The two errors pull oppositely and the negative one dominates, so the mean falls by one unit. Leaving it at 24 ignores both corrections; 25 applies them with reversed signs; 22 over-corrects. The corrected mean is 23.

13. (D) Each degree represents $\text{₹}24000 \div 360 = \text{₹}66.67$, but it is cleaner to use fractions of the circle. Rent = $(150 \div 360) \times 24000 = \text{₹}10000$. Savings = $(60 \div 360) \times 24000 = \text{₹}4000$. The difference = $10000 - 4000 = \text{₹}6000$ (equivalently the 90° gap is $(90 \div 360) \times 24000 = \text{₹}6000$). The value $\text{₹}4000$ is the Savings amount alone; $\text{₹}10000$ is the Rent amount alone; $\text{₹}90$ is the angle difference, not money. Rent exceeds Savings by $\text{₹}6000$.

14. (A) Adding 9 to each value raises the mean by 9, so the original mean = $31 - 9 = 22$. Since original mean $\times n$ = original total, $22 \times n = 110$, giving $n = 5$. The crucial step is undoing the +9 shift before using the total. Using 31 directly gives $110 \div 31 \approx 3.5$ (not whole); 22 is the recovered old mean mistaken for the count; 10 doubles the answer. The count is 5.

15. (C) Jar A = $3.5 \times 18 = 63$ marbles; Jar B = $5 \times 12 = 60$ marbles. So A has more, by $63 - 60 = 3$. Even though Jar B shows more symbols (5 vs $3\frac{1}{2}$), the larger scale on graph 1 makes A bigger — symbol counts cannot be compared directly across different scales. Comparing symbols alone (5 vs 3.5) wrongly favours B by 1.5; they are not equal; 18 is one symbol's value on graph 1, not the difference.

16. (B) Let the second group have k numbers. Totals: group1 = $3 \times 8 = 24$; group2 = $15k$; group3 = $4 \times 24 = 96$. Overall: $(24 + 15k + 96) \div (3 + k + 4) = 16$, i.e. $(120 + 15k) \div (k + 7) = 16$, so $120 + 15k = 16k + 112$, giving $k = 8$. Check: total = $24 + 120 + 96 = 240$ over 15 numbers = 16 ✓. Choosing 3 or 5 fails the equation; 2 under-counts. The second group has 8 numbers.

17. (C) Let Class Q have q students. $(40 \times 65 + 85q) \div (40 + q) = 77$, i.e. $(2600 + 85q) \div (40 + q) = 77$, so $2600 + 85q = 3080 + 77q$, giving $8q = 480$, $q = 60$. Check: total = $2600 + 5100 = 7700$ over 100 students = 77 ✓. Since 77 is closer to 85 than to 65 (gap 8 vs 12), Q must be the larger group, consistent with $60 > 40$. The values 20, 40 and 30 all break the balance equation. Class Q has 60 students.

18. (D) For eight sorted values the median is the average of the 4th and 5th values, a and b , so $(a + b) \div 2 = 17$, i.e. $a + b = 34$. The list is strictly increasing, so positions 4 and 5 must satisfy $12 < a < b < 21$. The three valid pairs (14, 20), (15, 19) and (16, 18) all sum to 34 AND obey $12 < a < b < 21$ strictly, so each can slot in between 12 and 21. The pair (18, 16) also sums to 34 but violates $a < b$ ($18 > 16$), so it cannot sit at positions 4 and 5 in increasing order — it is the unique impossible pair. The trap is checking only the sum while ignoring the strict ordering.

19. (B) Seven numbers total $7 \times 14 = 98$. The six known values sum to $5 + 8 + 14 + 18 + 18 + 24 = 87$, so $x = 98 - 87 = 11$. Check: the list becomes 5, 8, 11, 14, 18, 18, 24, still increasing ($8 < 11 < 14$) ✓; 18 appears twice and every other value once, so 18 is the unique mode ✓; and the mean is $98 \div 7 = 14$ ✓. The value 14 mistakes the mean for x ; 12 averages neighbours 8 and 14 wrongly; 9 makes the sum only 96, so the mean would not be 14. The value of x is 11.

20. (C) Originally v appears four times (the unique mode) and every other value once. Adding three copies of w makes w appear $0 + 3 = 3$ times, still fewer than v 's four. So v remains the single most frequent value: the set stays unimodal at v . To tie, w would need a fourth copy. Claiming w becomes the mode, that both tie, or that the mode vanishes all overstate what three copies achieve against v 's four.

21. (D) Children: $90 \rightarrow 126$, a rise of 36. Adults: $150 \rightarrow 132$, a fall of 18. So Children grew and Adults declined. Comparing 2024 values, Adults $132 >$ Children 126, so Adults still lead despite shrinking. Tall bars do not mean growth (Adults fell); Children did not overtake Adults ($126 < 132$); and Adults did not grow at all, so cannot have grown faster. Only the first statement is fully correct.

22. (B) There are $5 + 3 + 2 = 10$ equally likely outcomes. The number of green balls is 3, so $P(\text{green}) = 3/10$, and $P(\text{not green}) = 1 - 3/10 = 7/10$ (equivalently the 5 red + 2 blue = 7 non-green balls over 10). The value $3/10$ is $P(\text{green})$ itself; $2/10 = 1/5$ is $P(\text{blue})$; $1/2$ ignores the counts. The probability of not green is $7/10$.

23. (C) For five sorted values the median is the 3rd value. Only the two largest (positions 4 and 5) and the smallest (position 1) were changed, leaving the 3rd value untouched, so the median stays 15. The median resists changes confined to the extremes. Adding 8 (giving 23) or subtracting 5 (giving 10) wrongly assumes an edited value reaches the middle; 20 invents a shift that never touches position 3.

24. (B) Let Q have q numbers and P have $3q$. Total = $3q \times 24 + q \times 60 = 72q + 60q = 132q$ over $3q + q = 4q$ numbers, mean = $132q \div 4q = 33$. Because the lower-mean group P is three times as large, the combined mean leans strongly toward 24, landing at 33. The naive average of 24 and 60 is 42 (ignores the sizes); 36 under-weights P; 28 over-weights P. The combined mean is 33.

25. (A) Original total = $6 \times 25 = 150$; new total = $6 \times 29 = 174$, an increase of 24. Replacing 12 by w changes the total by $(w - 12)$, so $w - 12 = 24$, giving $w = 36$. The mean rose by 4 over 6 numbers, i.e. a total rise of 24 carried entirely by this swap. 29 copies the new mean; 24 is the total change, not w ; $41 = 29 + 12$ adds the old value instead of solving $w - 12 = 24$.

26. (D) Recorded total = $10 \times 60 = 600$. First error: 59 should be 95, add 36. Second error: 82 should be 28, subtract 54. Net correction = $+36 - 54 = -18$. True total = $600 - 18 = 582$, mean = $582 \div 10 = 58.2$. The two errors pull oppositely and the negative one dominates, so the mean falls slightly. Leaving it at 60 ignores both corrections; 61.8 applies them with reversed signs; 57 over-corrects. The corrected mean is 58.2.

27. (A) For seven sorted values the median is the 4th value, 30. Removing it leaves the original 1st, 2nd, 3rd, 5th, 6th, 7th values. The new median of six values is the average of the new 3rd and 4th positions, which are the original 3rd and 5th values — the immediate neighbours of 30. So the new median is their average, not 30 (which is gone) and not just one neighbour.

28. (C) Initially 6 appears 5 times and 9 appears twice. Adding three 9s makes 9 appear $2 + 3 = 5$ times. Adding one more 6 makes 6 appear $5 + 1 = 6$ times. So 6 (6 times) beats 9 (5 times) and stays the unique mode. The extra 6 is decisive — without it there would be a tie at 5. Choosing 9 ignores the added 6; the tie claim ignores 6 reaching 6; 'no mode' is false.

29. (C) Eight numbers total $8 \times 30 = 240$. The three added values sum to $30 + 10 + 50 = 90$, and $90 \div 3 = 30$ — exactly the existing mean. New total = $240 + 90 = 330$ over 11 numbers, mean = $330 \div 11 = 30$. Adding values whose own mean equals the current mean leaves the mean unchanged. 28 wrongly assumes the 10 drags it down; 31.8 uses a wrong count; 33.3 averages only the additions. The mean stays 30.

30. (C) With the axis starting at 500, only the parts above 500 are visible: X shows 30 units, Y shows 10 units, giving the 3-to-1 look. But the true values are 530 and 510, whose ratio is $530 \div 510 \approx 1.04$ — nearly equal, differing by just 20. The threefold appearance is an artefact of the cut axis. X is not three times Y, Y is clearly 510 (not zero), and the graph is readable once the axis start is noted.

31. (B) A linear transform $\text{new} = 5 \times \text{old} - 12$ applies directly to the mean: new mean = $5 \times 14 - 12 = 70 - 12 = 58$. The multiplier scales the mean and the constant shifts it.

Stopping after multiplying gives 70; leaving the mean unchanged ignores the transform; 22 mishandles the constant (e.g. $14 + 8$). The new mean is 58.

32. (B) Totals: $X = 45 + 30 = 75$; $Y = 18 + 52 = 70$, so Library X is larger overall ($75 > 70$). Gaps between the two languages: $X |45 - 30| = 15$; $Y |18 - 52| = 34$, so Y has the larger gap. The correct statement therefore says X is larger overall while Y has the bigger gap. Claiming Y is larger reverses the totals; the totals are not equal ($75 \neq 70$); and X does not have the larger gap ($15 < 34$).

33. (D) The 8 equally likely outcomes are 1, 2, 3, 4, 5, 6, 7, 8. The factors of 6 within this range are 1, 2, 3 and 6 — four favourable outcomes. So $P = 4/8 = 1/2$. Counting only 2, 3, 6 (forgetting 1 is a factor) gives $3/8$; counting only two of them gives $1/4$; $5/8$ over-counts. The probability is $1/2$.

34. (D) First three total $3 \times 16 = 48$; the five total $5 \times 24 = 120$, so the eight known numbers total $48 + 120 = 168$. Including x makes nine numbers with mean 22, so the total $= 9 \times 22 = 198$, giving $x = 198 - 168 = 30$. The eight-number total (168) is the key intermediate. 22 copies the final mean; 168 is the eight-number total, not x; 18 mishandles the arithmetic. The value of x is 30.

35. (A) Increasing the largest value by $40 - 22 = 18$ raises the total by 18, so the mean rises by $18 \div 6 = 3$, to 17. The median of six numbers is the average of the 3rd and 4th sorted values; the largest sits at position 6, so pushing it further out cannot touch positions 3 or 4 — the median stays 12. This is the median's robustness to a single extreme: 'both increase' ignores it, 'mean stays 14' ignores the changed total, and 'neither changes' ignores both effects.

36. (D) Totals: Drama $21 + 13 = 34$; Robotics $8 + 30 = 38$, so Robotics is more popular. Gaps: Drama $|21 - 13| = 8$; Robotics $|8 - 30| = 22$, so Robotics also has the larger gap. Both halves are correct. Saying Drama is more popular reverses the totals; pairing Robotics-popular with Drama-bigger-gap mixes the comparisons; the totals are not equal ($34 \neq 38$).

37. (B) Total change $= (3 - 7) + (3 - 9) = -4 - 6 = -10$, so the new total drops by 10; the mean falls by $10 \div 6 \approx 1.67$, from 18 to about 16.33. For the median of six values (average of the 3rd and 4th sorted values): the two smallest occupy positions 1 and 2, so lowering them does not touch positions 3 and 4 — the median stays 16. So the mean falls while the median holds at 16. 'Mean stays 18' ignores the lowered total; 'both fall by 2' wrongly moves the median; 'median falls to 12' is false because positions 3 and 4 never changed.

38. (A) Mean: the total changes by $+9 + 9 - 9 - 9 = 0$, so the mean stays 22. Median: only the two largest (positions 4 and 5) and two smallest (positions 1 and 2) changed, leaving the 3rd (middle) value untouched, so the median stays 22. Both are unchanged. The

balanced ± 9 changes cancel for the mean, and the median is unaffected because the centre never moved; every other option ignores one of these facts.

39. (D) The total increase is $2 + 4 + 6 + 8 + 10 = 30$. The original total was $5 \times 20 = 100$, so the new total is $100 + 30 = 130$ over 5 numbers, mean $= 130 \div 5 = 26$ (a rise of 6). Leaving the mean at 20 ignores the additions; 30 adds the increase to the mean directly; 22 adds only 2. The clean result is 26.

40. (D) Range = largest – smallest = 60. The largest drops by 15 $\rightarrow$ (largest – 15); the smallest rises by 9 $\rightarrow$ (smallest + 9). New range = (largest – 15) – (smallest + 9) = (largest – smallest) – 24 = $60 - 24 = 36$. Both ends move inward, shrinking the spread by $15 + 9 = 24$. Keeping 60 ignores the changes; 84 adds 24 instead of subtracting; 45 subtracts only one shift. The new range is 36.

41. (B) Recorded total = $16 \times 35 = 560$. The entry was $87 - 23 = 64$ too low, so the true total = $560 + 64 = 624$, mean $= 624 \div 16 = 39$. The error of 64 spread over 16 values raises the mean by 4, to 39. Leaving it at 35 ignores the correction; 37 spreads only half the error; 43 adds 8 instead of 4. The corrected mean is 39.

42. (A) Originally six values: median = average of 3rd and 4th $= (18 + 22) \div 2 = 20$. Remove 10, leaving five values 14, 18, 22, 26, 30: the median is now the 3rd value = 22. So the median rises from 20 to 22. It does not stay 20 (the count changed parity); it does not fall (a small value was removed, shifting the centre up); 24 is the average of 22 and 26, not the new middle. The median rises to 22.

43. (A) Bar heights in cm are 4, 5, 6, 5, 5; their mean height $= (4 + 5 + 6 + 5 + 5) \div 5 = 25 \div 5 = 5$ cm. Because the scale is linear, the bar for the mean mark also has the mean height, 5 cm. (Check: marks 24, 30, 36, 30, 30, mean 30, and $30 \div 6 = 5$ cm.) The value 30 is the mean in marks, not cm; 25 is the total height; 1 confuses dividing height by the scale. The bar is 5 cm tall.

44. (A) Totals: $A = 5 \times 12 = 60$; $B = 3 \times 20 = 60$; $C = 2 \times 30 = 60$. Grand total = $60 + 60 + 60 = 180$ over $5 + 3 + 2 = 10$ numbers, mean $= 180 \div 10 = 18$. Each group happens to contribute 60, but the means must be weighted by group size, not averaged. Naively averaging 12, 20, 30 gives about 20.7; 62 mistakes the change; 15 under-counts. The combined mean is 18.

45. (B) Sorted already. Mode = 11 (three times). Median = 5th value = 14. Mean $= (7+11+11+11+14+18+20+21+22) \div 9 = 135 \div 9 = 15$. So mean + mode – $2 \times$ median $= 15 + 11 - 2 \times 14 = 26 - 28 = -2$. Mislocating the median as the 4th value (11) gives 0; reading it as the 6th value (18) shifts the sign to +; 15 is just the mean. The expression equals –2.

46. (D) Sorted: 14, 14, 14, 18, 22, 22, 26. Mode = 14 (three times). Median = 4th value = 18. Mean $= (14+14+14+18+22+22+26) \div 7 = 130 \div 7 \approx 18.6$. So 14 (mode) < 18 (median) < 18.6 (mean): mode < median < mean. The cluster of low modal 14s puts the mode lowest,

while the high values 22, 22, 26 lift the mean just above the median. The other orderings misplace the mean or wrongly equate the three.

47. (B) The transform $\text{new} = (\text{old} \div 2) + 12$ applies to the mean: $\text{new mean} = (40 \div 2) + 12 = 20 + 12 = 32$. Halving alone gives 20 (forgets the +12); adding 12 to the original mean before halving, or other mis-sequencing, gives 52 or 26. The correct new mean is 32.

48. (A) Forest M = $3 \times 25 = 75$ trees; Forest N = $4 \times 20 = 80$ trees. So N has more, by $80 - 75 = 5$. Even though the scales differ, converting first shows N edges ahead. Comparing symbol counts alone (4 vs 3) wrongly favours N by 1; M does not lead; they are not equal. Forest N is greater by 5 trees.

49. (B) Recorded total = $14 \times 50 = 700$. The entry was $96 - 18 = 78$ too HIGH, so the true total = $700 - 78 = 622$, mean = $622 \div 14 \approx 44.4$. The over-entry by 78 spread over 14 values lowers the mean by about 5.6. Leaving it at 50 ignores the correction; 55.6 adds instead of subtracts; 47 corrects only partly. The corrected mean is about 44.4.

50. (D) For nine sorted values the median is the 5th value. The changes touch only positions 1–4 and 6–9; position 5 is never altered and the order is preserved (the gaps only widen), so the median stays exactly 50. The symmetric ± 30 changes spread the data out far more, but the centre value does not move. Adding 30 (giving 80) or subtracting 30 (giving 20) wrongly assumes the 5th value was among those changed; 65 has no basis. The median is unchanged at 50.

51. (C) After 12 innings his total is $12 \times 38 = 456$. To have a mean of 42 over 14 innings, his total must be $14 \times 42 = 588$. So the next two innings together must add $588 - 456 = 132$ runs. Scoring 84 ($= 2 \times 42$, the target mean per innings) gives a total of 540 over 14 = 38.6, not 42. 120 and 66 mis-handle the totals. He needs 132 runs across the two innings.

52. (D) Total of all eight = $8 \times 25 = 200$. Four numbers each equal to the mean contribute $4 \times 25 = 100$. So the remaining four sum to $200 - 100 = 100$. The whole total 200 is for all eight, not four; 25 is one value; 50 forgets half of the mean-valued numbers. The other four sum to 100.

53. (B) Total = $5 \times 40 = 200$. Removing the smallest and largest: $200 - 12 - 84 = 104$ is shared equally by the three middle numbers, so each = $104 \div 3 \approx 34.7$. The value 104 is their combined total, not each one; 40 copies the overall mean; 52 wrongly divides 104 by 2. Each middle number is about 34.7.

54. (C) The angles must add to $140 + 100 + 80 + 40 = 360^\circ$ ✓. Football's share = $(100 \div 360) \times 720 = 720 \times (5 \div 18) = 200$ students. (Each degree is $720 \div 360 = 2$ students, so $100^\circ \rightarrow 200$.) The value 100 is Football's angle, not its count; 280 is Cricket's count (140×2); 160 mis-multiplies. Football was chosen by 200 students.

55. (A) Both shops total ₹120 thousand, so seasonal totals match — but equal totals do not mean equal quarters: P's values are 40, 70, 10 while Q's are 30, 30, 60, a clearly different period-by-period pattern. So 'identical' is wrong even though totals agree. The totals do NOT differ (both 120); equal totals do not make the data identical; and a single largest quarter (Q's 60) does not mean Q sold more overall since the totals tie.

56. (C) Original total = $9 \times 22 = 198$. New mean 20 over 9 numbers needs a total of $9 \times 20 = 180$, a drop of 18. Replacing 46 by v changes the total by $(v - 46)$, so $v - 46 = -18$, giving $v = 28$. The drop of 18 is the change in total, not the value; 18 is that change; 46 is the original value; 20 is the new mean. The replacement is 28.

57. (C) Adding 130 (well above 50) raises the total, so the mean rises above 50. The mode is the most frequent value; one new 130 cannot outnumber the many 50s, so the mode stays 50. The median, being position-based, shifts only slightly toward the upper-middle (it cannot leap to 130, an extreme). So the mean is most affected, the mode is unmoved, and the median barely moves — the classic outlier pattern. The mode is defined by frequency, not size; the three do not move together; and adding an extreme value certainly changes the mean.

58. (D) List with $y = 13$: sorted 5, 9, 13, 13, 13, 17. Mode = 13 (three times) ✓. Median = average of 3rd and 4th = $(13 + 13) \div 2 = 13$ ✓. So $y = 13$ keeps both. For $y = 11$: sorted 5, 9, 11, 13, 13, 17, median = $(11 + 13) \div 2 = 12$, not 13, so the median fails. For $y = 5$: sorted 5, 5, 9, 13, 13, 17, median = $(9 + 13) \div 2 = 11$, not 13. $y = 18$: sorted 5, 9, 13, 13, 17, 18, median = $(13 + 13) \div 2 = 13$ holds but it need not be exactly 18 — $y = 13$ also works, and the claim 'must be exactly 18' is false. Only $y = 13$ is guaranteed to keep both.

59. (A) With the axis starting at 1000, the visible parts are 40 and 10 units, so X's bar looks about four times Y's. But the true values are 1040 and 1010, differing by just 30 — about 3% — so the cities are nearly the same size. The truncated axis manufactured the illusion. X is not four times Y, Y is plainly 1010 (not zero), and the graph does not prove a big difference once the axis start is read honestly.

60. (C) The six equally likely outcomes are 1, 2, 3, 4, 5, 6. Even numbers: 2, 4, 6. Numbers greater than 4: 5, 6. The outcomes in A OR B (listing each once, since 6 is shared) are 2, 4, 5, 6 — four favourable outcomes. So $P = 4/6$ (which simplifies to $2/3$). Counting only the evens gives $3/6$; double-counting the shared 6 inflates it to $5/6$; $2/6$ counts only 'greater than 4'. The probability is $4/6$.